Final Report Of Robotics



by

Yuwei Zhao[1], Haohao Qi[2], Yilin Zhang[3]

In contributions of

[1] *Data Processing & Experiment Implementation*
[2] *Initial Draft & Images Collection*
[3] *Paper Writing & Experiment Implementation*

ROBOTICS

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

HERIOT-WATT UNIVERSITY

June 2025

Abstract

In accordance with the Academic Regulations, the thesis must contain an abstract preferably not exceeding 200 words, bound in to precede the thesis. The abstract should appear on its own, on a single page. The format should be the same as that of the main text. The abstract should provide a synopsis of the thesis and shall state clearly the nature and scope of the research undertaken and of the contribution made to the knowledge of the subject treated. There should be a brief statement of the method of investigation where appropriate, an outline of the major divisions or principal arguments of the work and a summary of any conclusions reached. The abstract must follow the Title Page.

Table of Contents

1 Introduction **1**
- 1.1 Four aims of the laboratory . . . 1
- 1.2 Results summary . . . 1

2 Experiments **2**
- 2.1 Light Seeking . . . 2
  - 2.1.1 Theory and Technical Background . . . 2
  - 2.1.2 Method . . . 2
  - 2.1.3 Results . . . 2
- 2.2 Line following . . . 2
  - 2.2.1 Theory and Technical Background . . . 2
  - 2.2.2 Method . . . 3
  - 2.2.3 Results . . . 3
- 2.3 Obstacle avoiding . . . 3
  - 2.3.1 Theory and Technical Background . . . 3
  - 2.3.2 Method . . . 3
  - 2.3.3 Results . . . 4
- 2.4 Environment Measurement . . . 4
  - 2.4.1 Theory and Technical Background . . . 4
  - 2.4.2 Method . . . 4
  - 2.4.3 Results . . . 4

3 Conclusions **5**

A Appendix **6**
- A.1 Contribution Statement . . . 6
- A.2 Source Code . . . 6

Chapter 1

Introduction

1.1 Four aims of the laboratory

The aim of this laboratory is to build our own robotic car. It is named as "**Super Ultra Hyper Mega Advanced Senior Superior Elite Lobster**", short as "**Lobster**". We have done **4** contributions to Lobster, which are recorded in lab3-light seeking, lab5-line following, lab7-Obstacle avoiding and lab8-environment measurement. Thus, Lobster could have the following functionalities:

1. Follow the light on the floor and turn its direction to wherever the light is.
2. Follow a 1 to 3 cm wide black line on the floor. The line could be straight, bent or quarter.
3. Go straight and avoid obstacles. When reaches obstacles such as walls, Lobster will turn to another direction to avoid crashing.
4. Measure temperature, humidity and atmospheric pressure, print them and set a motor speed dependently.

If Lobster has installed light seeking module, then it cannot track lines any more, same as the other two functionalities. In short, only one of these four modules works at a time.

1.2 Results summary

Basic approaches of the four experiment are completed. We used two light-dependent resistors (LDR) to receive light change. The car can recognize white light in the front within 15 cm.

Chapter 2

Experiments

2.1 Light Seeking

2.1.1 Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

2.1.2 Method

Abridge	Full Name
AAA	AAAAAAA
BBB	BBBBBBBBB
CCC	CCCCC

Keywords: “Key”, Key, Key!
Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

2.1.3 Results

Add all appropriate figures and analysis. No formal word count for this section, however, please keep text as concise as possible.

2.2 Line following

2.2.1 Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each fig-

ure and each equation may reduce the number of words you use.

2.2.2 Method

Abridge	Full Name
AAA	AAAAAAA
BBB	BBBBBBBBB
CCC	CCCCC

Keywords: “Key”, Key, Key!

Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

2.2.3 Results

Add all appropriate figures and analysis. No formal word count for this section, however, please keep text as concise as possible.

2.3 Obstacle avoiding

2.3.1 Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

2.3.2 Method

Abridge	Full Name
AAA	AAAAAAA
BBB	BBBBBBBBB
CCC	CCCCC

Keywords: “Key”, Key, Key!

Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

2.3.3 Results

Add all appropriate figures and analysis. No formal word count for this section, however, please keep text as concise as possible.

2.4 Environment Measurement

2.4.1 Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

2.4.2 Method

Abridge	Full Name
AAA	AAAAAAA
BBB	BBBBBBBBB
CCC	CCCCC

Keywords: “Key”, Key, Key!
Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

2.4.3 Results

Add all appropriate figures and analysis. No formal word count for this section, however, please keep text as concise as possible.

Chapter 3

Conclusions

Please give the conclusions from the characterisations including a summary of the results collected.

Appendix A

Appendix

Appendices, labelled A, B etc., should be treated as additional chapters and should normally follow the main text. Appendices may consist of supporting material of considerable length or of lists, documents, commentaries, tables or other evidence that if included in the main text, would interrupt its flow. The style of appendices must be consistent with the style of the main text. Long appendices may be divided into sections, labelled as Appendix A.1 etc., with corresponding subsection numbering, which must be entered in the table of contents. Alternatively, short appendices may be attached to individual chapters, as an extra section with a heading of style 3.7 Appendix

A.1 Contribution Statement

The contributions of each team member are as follows:
– Yuwei Zhao: Responsible for data processing and experiment implementation.
– Yilin Zhang: Contributed to the initial draft and collected supporting images.
– Haohao Qi: Wrote the final report and participated in experiment implementation.

A.2 Source Code

```
print("Hello World")
```